

Communications and Cybersecurity

Presentation

Outline

- Presentation
- Objectives & Topics
- Plan
- Evaluation
- Question & Comments

Presentation

- What are my expectations with this course ?

@Whomai

- Bruno Sousa, bmsousa@dei.uc.pt
- Member of NCS group

@ReachOut

- On Signal app
- Presential D3.13 on Monday at 14h-16h, Wednesday at 09h-11h



Presentation / Professors

- T and PL1
 - Bruno Sousa, bmsousa@dei.uc.pt
- PL2
 - Nuno Simões, nuno.simo@uc.pt
 - Reach Out (agree with professor):
 - Online Wednesday 12h-14h



Objectives & Topics

Knowledge of the principles, algorithms, techniques and approaches to the security of communications and systems.

- Symmetric encryption
- Asymmetric encryption
- Cybersecurity on the Internet
- Key Management
- Policies and Procedures
- Wireless network security
- Cloud Security
- Network and transport layer security
- Post Quantum Cryptography

Plan

Date	#	T	PL
2026-09-07	0	Welcome Week	
2026-09-14	1	Introduction, Cybersecurity concepts, Security of communications, types of attacks	Exercise 1 - Sniffing of web traffic
2026-09-21	2	OSI Model and security in layers, introduction to cryptography and brief historical perspective.	Exercise 2 - Working with TLS
2026-09-28	3	Symmetric Cryptography, Stream cipher, and block cipher, permutation networks and Feistel networks. Symmetric Cryptography - DES, AES and other algorithms (Chacha20, Camellia). Selection principles.	Exercise 2 - Working with TLS
2026-10-05		No theoretical Classes (Latada)	Support to Assignment (PL1 - optional in the PL2 slot)
2026-10-12	4	Hashing and Message authentication Codes (MAC)	Defenses
2026-10-19	5	Asymmetric Cryptography - Introduction, principles, RSA, Elliptic Curves	Exercise 3 - PKI with OpenSSL
2026-10-26	6	Asymmetric Cryptography Post Quantum + Certificates + PKI	Exercise 4 - Hashing for integrity control (BD)
2026-11-02	7	Applications	Exercise 5 - DNS + DNSSEC
2026-11-09	8	Blockchain + DNS + DNSSEC	Exercise 6 - Pretty Good Privacy and Secure Shell (SSH)
2026-11-16	9	PGP and SSH Protocols Security in Communications - Wireless Networks I	Exercise 7 - PGP + EAP Part I
2026-11-23	10	Security in Communications - Wireless Networks II	Exercise 7 - PGP + EAP Part II
2026-11-30	11	Security in Communications - Transport (TLS, Noise)	Exercise 8 - Firewalls Exercise 9 - SIEMs
2026-12-07	12	Security in Communications - Firewalls, SIEMS	Support to Assignment
2026-12-14	13	Advanced Topics, homomorphic encryption, Zero Trust	Defenses

Red & Blue Teams

- Collaborative approach for security.
- **Blue team** is the defending team, implementing security measures, securing systems, infrastructures.
- **Red Team** is the team performing attacks, emulating the behaviour of real hackers so that Blue Team can improve.



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[Definition from NIST](#)

Evaluation

- Evaluation considers:
 - Resolution of Exercises in PLs (5 values)
 - 9 Exercises suggested
 - 2 Practical Assignments (9 values)
 - First PA: 4 values
 - Final PA: 5 values
 - Exam (6 values)

Evaluation (details)

- Evaluation considers:
 - Resolution of Exercises in PLs (5 values)
 - 9 Exercises suggested
 - Evaluation in groups of 2 (need to state roles in each work):
 - One member as Blue Team – Defence
 - One member as Red Team – Attack
 - Evaluation criteria (regular students):
 - 80% of the evaluation gathered during the PL class – Participation in the discussion at the end of the class (10 minutes)
 - 20% considering the written report (to be delivered after class)
 - Evaluation criteria (working students / remote students):
 - 80% evaluation from the defence (e.g., answer questions)
 - 20% considering the written report

Evaluation (details)

- Evaluation considers:
 - Resolution of Exercises in PLs (5 values)
 - 2 Practical Assignments (9 values)
 - First PA: 4 values To be delivered on 11/November/2026
 - Final PA: 5 values To be delivered on 13/December/2026
 - Exam (6 values)
 - Without consultation

Question & Comments

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